

Effect of Ti content on the microstructure and mechanical properties of $\text{Ti}_x\text{ZrNbMoV}$ lightweight refractory high-entropy alloys

Zhenghang Cao^a, Huaican Chen^b, Wei Wang^a, Jianwen Guo^a, Zhenzhong Sun^a, Yang Hai^{a,*}, Wen Yin^{b,*}

^a School of Mechanical Engineering, Dongguan University of Technology, Dongguan 523808, Guangdong Province, China

^b Institute of High Energy Physics, Chinese Academy of Science, Spallation Neutron Source Science Center, Dongguan 523803, China

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ABSTRACT

In this study, we prepared $\text{Ti}_x\text{ZrNbMoV}$ ($x = 1, 1.5$, and 2) refractory high-entropy alloys (RHEAs) with varying amounts of Ti by vacuum arc melting. The microstructures of RHEAs are investigated by neutron diffraction, electron backscatter diffraction and transmission electron microscope. The results showed that the $\text{Ti}_x\text{ZrNbMoV}$ alloys mainly comprise body centered cubic phases and hexagonal close packed phases and the average grain size of alloys decreases as the Ti element content increases. Room-temperature compressive tests showed that increasing the Ti content of the alloys improved their ductility, although their strength decreased. Compared with some previously reported RHEAs, the $\text{Ti}_x\text{ZrNbMoV}$ RHEAs prepared in this study possessed better comprehensive mechanical properties, with the specific yield strength of the TiZrNbMoV alloy being as high as $312 \text{ kPa}\cdot\text{m}^3/\text{kg}$. Finally, the deformation behavior of $\text{Ti}_x\text{ZrNbMoV}$ RHEAs was studied by nanoindentation.

1. Introduction

Refractory high-entropy alloys (RHEAs), a type of multi-component alloy composed mainly of refractory metal elements, were first reported by Senkov [1] in 2010. Their use is expected to solve problems associated with traditional alloys in industries such as the aerospace and nuclear industries. Compared with conventional single-principal element alloys, high-entropy alloys (HEAs) exhibit high strength [2,3], high-temperature resistance [4], wear resistance [5–7], and radiation resistance [8], and therefore, have great potential for various industrial applications. However, with their high strength, most previously reported RHEAs have a high density and poor ductility at room temperature. Some RHEAs also exhibit negligible plasticity [9]. In addition, RHEAs have insufficient specific yield strength (SYS), which severely restricts their industrial applications.

The element proportions and types of RHEAs are commonly adjusted to improve their mechanical properties. Wang [10] increased the content of V in the RHEAs that inhibit impurities and brittle compounds at grain boundaries, making the alloy prone to transgranular fracture, and thus improving their plasticity. Liu [11] prepared a series of RHEAs by adding Si elements, the yield strength increased from 1260 MPa without Si to maximum value of 2134 MPa, while the density decreased from

9.02 g/cm^3 to minimum value of 8.49 g/cm^3 . Sheikh [12] proposed that the ductility of RHEAs can be improved by adding Ti and Zr elements, and TiZrNbMoV alloy system is a potential RHEAs that possesses the characteristics of low density and good ductility. Huang [13] found that increasing the Ti content in an alloy can inhibit the formation of the detrimental V_2Zr phase and enhances the plasticity of RHEAs. In addition, Ti is beneficial to refine the grain and improve the plasticity of the alloys [14–16]. Wu [17] et al. obtained the $\text{TiZrNbMo}_{0.3}\text{V}_{0.3}$ RHEAs with the yield strength and plastic strain of 1312 MPa and $> 50\%$, respectively. Meanwhile the plasticity of RHEAs can be improved by adjusting the proportion of Mo and V in the TiZrNbMoV alloy system.

Most reported RHEAs have a single-phase body centered cubic (BCC) structure [17–19] because of the formation of solid-solution phases. Multiphase alloys usually have higher strength but too much second phase will seriously deteriorate the plasticity [20]. Therefore, it is expected to obtain RHEAs with good comprehensive mechanical properties by introducing and regulating the second phase in RHEAs.

In this study, we designed, prepared, and characterized $\text{Ti}_x\text{ZrNbMoV}$ RHEAs, that contained varying amounts of Ti ($x = 1, 1.5$, and 2). We investigated the effects of varying the Ti content on the microstructure and compressive properties of the synthesized alloys using neutron diffraction (ND), electron backscatter diffraction (EBSD), transmission

* Corresponding authors.

E-mail addresses: haiyang@dgut.edu.cn (Y. Hai), yinwen@ihep.ac.cn (W. Yin).

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electron microscope (TEM) and compressive tests. We also investigated the creep behavior of the three alloys using nanoindentation tests.

2. Materials and methods

$\text{Ti}_x\text{ZrNbMoV}$ ($x = 1, 1.5$ and 2) alloys were prepared by vacuum arc melting using Ti, Zr, Nb, Mo, and V raw materials with >99.99 wt% purity. The vacuum chamber was evacuated to 3×10^{-4} Pa and then flushed with Ar gas to 0.05 MPa to prevent oxidation of the alloys during the melting process. The evacuation and flushing steps were repeated three times. All the resulting ingots were remelted and flipped five times during the arc-melting process to ensure homogeneous distribution of the elements.

The microstructures of the $\text{Ti}_x\text{ZrNbMoV}$ ($x = 1, 1.5$ and 2) alloys were analyzed using ND, EBSD and TEM. The ND experiments were performed using the Multi-Physics Instrument at the China Spallation Neutron Source (CSNS). EBSD analyses were performed using an EDAX-TSL system and a Nordlys Max3 EBSD detector. TEM were performed in a FEI Tecnai G2 F20 microscope, operated at 200 Kv, and the TEM samples were prepared by ion-beam thinning.

Alloy specimens were prepared with dimensions of $\Phi 4 \text{ mm} \times 6 \text{ mm}$ from the as-cast alloy ingot using electric discharge machining. The room-temperature compressive tests were performed on a universal testing machine. The compressive strain rate was set to $1 \times 10^{-3}/\text{s}$ in all the mechanical property tests. The test was repeated three times for each alloy.

Nanoindentation tests (Nano Indenter G200) were conducted with a Berkovich-type indentation tip at room temperature. Before performing the nanoindentation creep tests, the surface of each sample was polished to smoothness and a loading rate of 2 mN/s and a load of 400 mN were maintained for 60 s. To ensure data reliability, four nanoindentation experiments were performed for each condition, with each indentation point located at least 50 μm apart.

3. Results and discussion

3.1. $\text{Ti}_x\text{ZrNbMoV}$ RHEAs microstructures

The ND patterns are shown in Fig. 1. BCC and hexagonal close packed (HCP) are seen in the $\text{Ti}_x\text{ZrNbMoV}$ ($x = 1, 1.5, 2$) alloys by refining ND data. This is consistent with the literature [20].

Fig. 2(a)–(c) show the phase analysis maps of the $\text{Ti}_x\text{ZrNbMoV}$ alloys obtained by EBSD. $\text{Ti}_x\text{ZrNbMoV}$ mainly exhibits a BCC phase and a few

HCP phase, and the HCP phase content decreases with the increases of Ti content. We observed that the proportion of the HCP phase decreased with increasing Ti content, indicating that a higher Ti content inhibited the formation of the HCP phase.

The orientation maps in Fig. 2(d)–(f) show that the grains of $\text{Ti}_x\text{ZrNbMoV}$ alloys have random orientations. Moreover, TiZrNbMoV and $\text{Ti}_{1.5}\text{ZrNbMoV}$ exhibit a typical dendritic microstructure, while the $\text{Ti}_2\text{ZrNbMoV}$ alloys exhibited an equiaxed dendritic microstructure. The structural change from a typical dendritic to an equiaxed dendritic microstructure with the increase of Ti content. This is because the increase of the undercooling of the alloy structure following the increasing of Ti content promotes the evolution of the alloy grain to equilibration [21]. In addition, some nanocrystals formed between and inside the dendritic structures of the alloys. The locations of the nanocrystals observed in Fig. 2(d)–(f) coexisted with the HCP-phase locations, indicating that some HCP-phase nanocrystals precipitated at the BCC phase grain boundary.

Fig. 3(a)–(c) show the grain-size distribution maps of the $\text{Ti}_x\text{ZrNbMoV}$ alloys. The TiZrNbMoV alloy was composed of coarse and fine grains with large differences in grain size. At higher Ti contents, the grain-size distribution became more uniform. The average grain sizes of the TiZrNbMoV , $\text{Ti}_{1.5}\text{ZrNbMoV}$, and $\text{Ti}_2\text{ZrNbMoV}$ alloys were approximately 81, 33, and 16 μm , respectively, indicating that the increase in Ti content led to a reduction grain size. That is consistent with the literatures [14–16]. That is because the Ti content of the alloys is conducive to the enrichment of elements ahead of the solid–liquid boundary, which can result in supercooling in the diffusion layer at the advancing solid/liquid interface [22]. Therefore, the crystal growth rate was limited, and grain refinement was achieved.

Fig. 4(a)–(c) show the grain boundary misorientation maps of the $\text{Ti}_x\text{ZrNbMoV}$ alloys. The colors blue, green, and red correspond to the high-angle grain boundary ($>15^\circ$), low-angle grain boundary (approximately 15° – 5°), and low-angle grain boundary (approximately 5° – 2°), respectively. Many high-angle grain boundaries (HAGBs) can be seen in the grain boundaries, while low-angle grain boundaries (LAGBs) can be seen around the HAGBs and inside the dendritic structure. These LAGBs might be formed by second-phase nanocrystals and the serious lattice distortion of RHEAs results in the formation of HAGBs.

Fig. 5(a)–(c) show the kernel average misorientation (KAM) maps of the $\text{Ti}_x\text{ZrNbMoV}$ alloys. The region with HCP-phase nanocrystals was in a high-strain state (suggesting high dislocation densities in this region [23]), while the other regions were in a low-strain state. The dislocations may have been caused by the high cooling rate owing to the small volume of the alloy samples prepared by vacuum arc melting [24].

The phase structure of the $\text{Ti}_x\text{ZrNbMoV}$ alloys were further analyzed by TEM. Fig. 6 shows the TEM bright field and corresponding selected area electron diffraction (SAED) images of $\text{Ti}_x\text{ZrNbMoV}$ alloys. The $\text{Ti}_x\text{ZrNbMoV}$ alloys showed both HCP, BCC1 and BCC2 phase. The chemical composition of different phases of $\text{Ti}_x\text{ZrNbMoV}$ alloys were given in Table 1 by TEM-EDS. According to TEM-EDS analysis, the HCP phase was enriched with Zr and depleted with Ti and Nb, and the BCC2 phase was depleted with Zr in $\text{Ti}_x\text{ZrNbMoV}$ alloys. The chemical composition of BCC1 phase was enriched with Zr in TiZrNbMoV and $\text{Ti}_2\text{ZrNbMoV}$ alloy. For $\text{Ti}_{1.5}\text{ZrNbMoV}$ alloy, the chemical composition of BCC1 phase is approaches to the nominal value.

3.2. Compressive properties of $\text{Ti}_x\text{ZrNbMoV}$ RHEAs

The VEC, a parameter used to estimate alloy ductility, can be calculated using the following equation [25]:

$$\text{VEC} = \sum_{i=1}^n c_i (\text{VEC})_i, \quad (1)$$

where c_i and $(\text{VEC})_i$ are the atomic percentage and VEC of the i th component, respectively.

The differences in atomic radius, δ , can be calculated using the

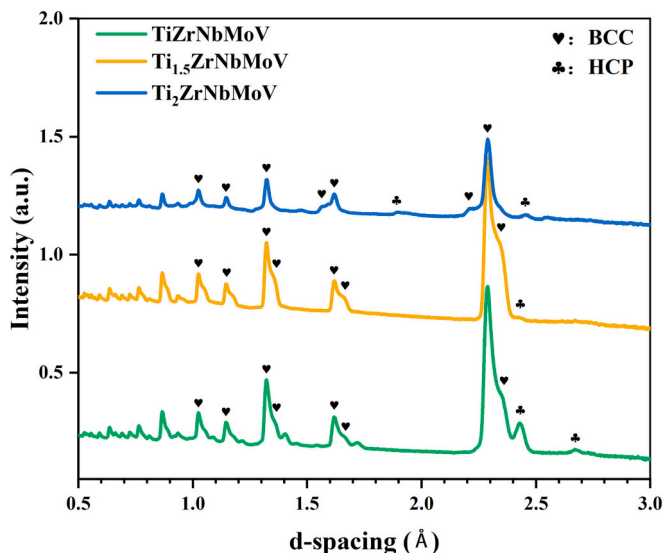


Fig. 1. ND patterns of $\text{Ti}_x\text{ZrNbMoV}$ alloys.

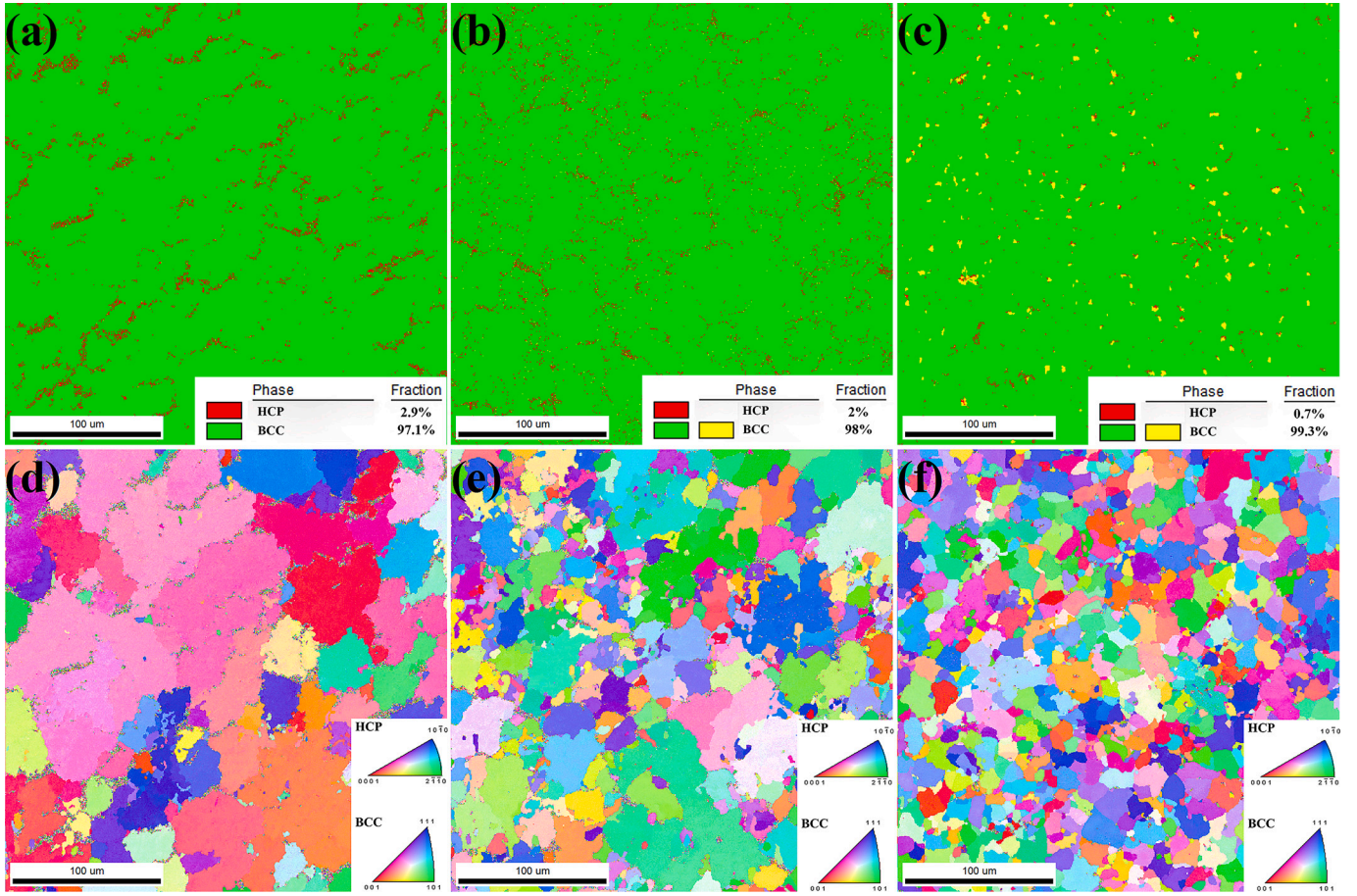


Fig. 2. EBSD phase analysis maps of RHEAs (a) TiZrNbMoV, (b) Ti_{1.5}ZrNbMoV, and (c) Ti₂ZrNbMoV and orientation maps of (d) TiZrNbMoV, (e) Ti_{1.5}ZrNbMoV, and (f) Ti₂ZrNbMoV.

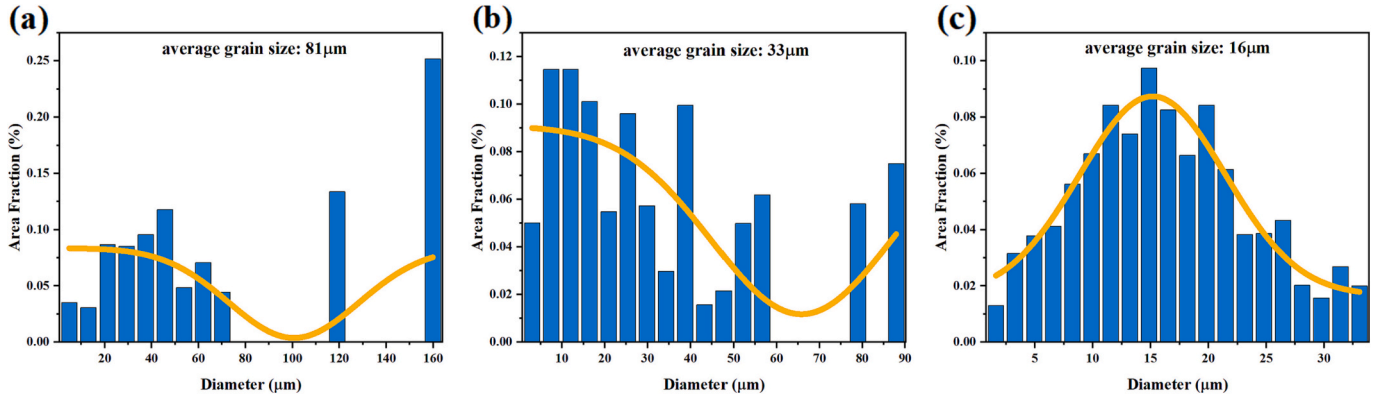


Fig. 3. Grain-size distribution map of (a) TiZrNbMoV, (b) Ti_{1.5}ZrNbMoV, and (c) Ti₂ZrNbMoV.

following equation [26]:

$$\delta = \sqrt{\sum_{i=1}^n c_i \left(1 - \frac{r_i}{\sum_{i=1}^n c_i r_i} \right)^2} \times 100\%, \quad (2)$$

where c_i and r_i are the atomic percentage and radius of the i th component, respectively. Table 2 lists the VEC and δ values for the Ti_xZrNbMoV alloys.

Fig. 7(a) and 7(b) show the engineering stress–strain curves, yield strengths, and fracture strains of the Ti_xZrNbMoV alloys. The figures

show that a higher Ti content leads to a higher fracture strain and reduced yield strength.

The three Ti_xZrNbMoV alloys possess high strength owing to the solid solution and second-phase strengthening mechanisms. The elements constituting the RHEAs have large atomic radii differences, and their random occupation within the lattice causes severe lattice distortion that results in high shear stress of dislocation slip, making dislocation slip difficult to achieve. The decrease in δ values with increasing Ti content indicates a reduction in the degree of lattice distortion that is one of the reasons for the decrease in the alloy strength.

In addition to the solid-solution strengthening mechanism, the

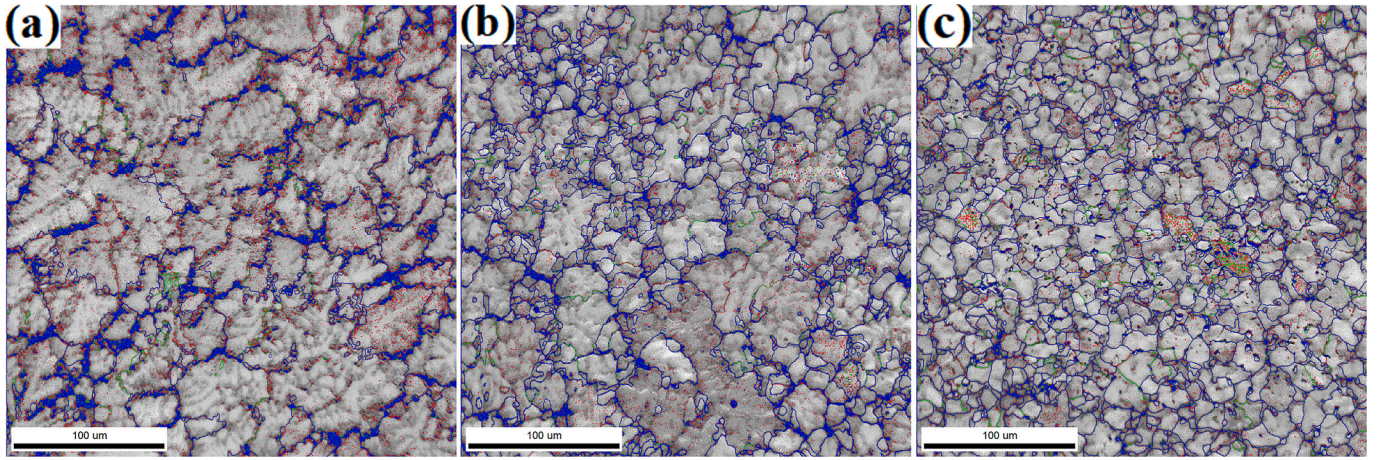


Fig. 4. Grain boundary misorientation map of (a) TiZrNbMoV, (b) Ti_{1.5}ZrNbMoV, and (c) Ti₂ZrNbMoV.

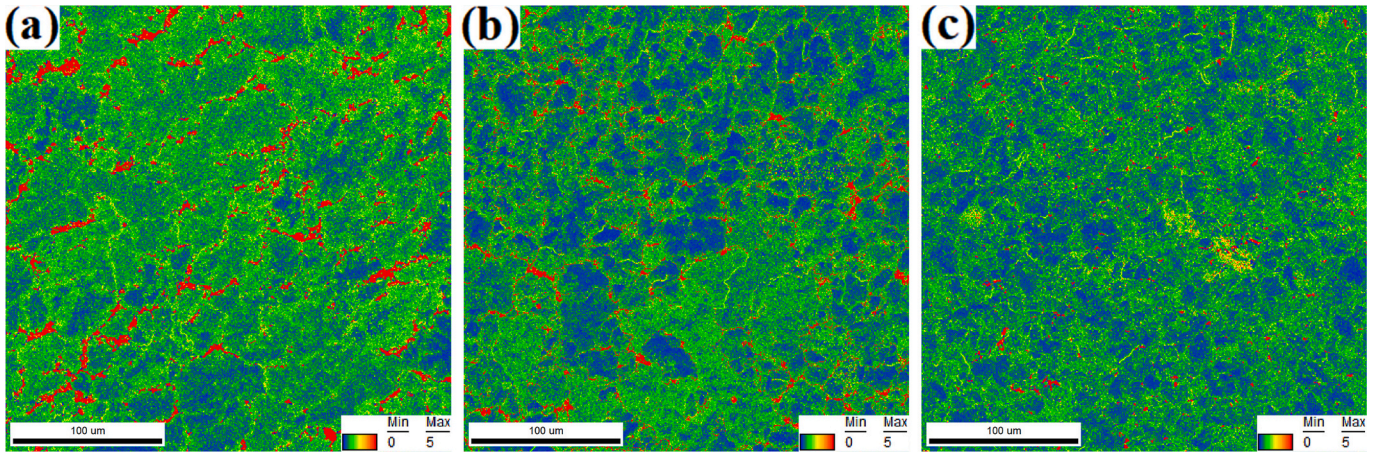


Fig. 5. KAM map of (a) TiZrNbMoV, (b) Ti_{1.5}ZrNbMoV, and (c) Ti₂ZrNbMoV.

second-phase strengthening mechanism can affect the mechanical properties of the Ti_xZrNbMoV alloys. Li [27] used the models proposed by Senkov [28] to estimate the yield stress of RHEAs. And found differences between the theoretical and calculated values attributed to second-phase strengthening. Nanocrystals of the second phase at grain boundaries can cause dislocation accumulation and effectively prevent dislocation slip, thereby improving the yield strength of an alloy. The fraction of the second phase decreased as the Ti content increased, indicating that the second-phase strengthening effect weakened. This is another important reason for the reduced strength of the Ti_xZrNbMoV alloys.

The improvement in alloy ductility resulted from grain refinement caused by an increase in the Ti content. Finer grains can make the plastic deformation of the alloy uniform, preventing the occurrence and propagation of cracks caused by excessive local stress concentration. Moreover, increasing the Ti content reduces the VEC values of the three Ti_xZrNbMoV alloys. For BCC alloy, (100) is the direction of the worst plasticity, if shear instability occurs before the ideal tensile stress threshold, dislocation nucleation will be activated, and the alloy will occur plastic deformation [29]. VEC reduction is beneficial for increasing the ductility of RHEAs because it lowers the Fermi level of the alloys, making them more prone to shear instability and activation of the dislocation nucleation energy during plastic deformation [17,25,29].

The specific yield strength (SYS), the ratio of the yield strength to the density of an alloy, is an important reference parameter for industrial metallic structural materials. The estimated density of RHEAs, ρ_{mix} , of a disordered solid solution is given by [4]

$$\rho_{mix} = \frac{\sum c_i A_i}{\sum \frac{c_i A_i}{\rho_i}}, \quad (3)$$

where c_i , A_i , and ρ_i are the atomic percentage, weight, and density of the i th element, respectively. Table 2 lists the densities and SYS of the Ti_xZrNbMoV alloys. The density of the Ti_xZrNbMoV alloys decreased with increasing Ti content. The Ti₂ZrNbMoV alloy possessed the lowest density (6.69 g/cm³), while the TiZrNbMoV alloy had the highest SYS value (~312 kPa·m³/kg). As shown in Fig. 8, the Ti_xZrNbMoV alloys possessed excellent SYS, low density, and acceptable ductility compared to most previously reported RHEAs [4,30–35].

3.3. Nanoindentation creep of Ti_xZrNbMoV RHEAs

HEAs exhibit slow diffusion and lattice distortion effects, which have significant effect on plastic deformation of the alloys. And the plastic deformation at the submicron scale can be obtained by nanoindentation. Plastic deformation caused by nanoindentation occurred at the submicron scale. Nanoindentation creep behavior is considered to have a strong size effect, and the value of the stress exponent varies significantly at different creep depths, especially below 100 nm [36,37]. To avoid the size effect, we used a high load of 400 mN and a loading rate of 2 mN/s to increase the depth of the nanoindentation to >1600 nm in all experiments. We used the creep stress exponent n and strain rate sensitivity m during the steady-state creep stage to evaluate the creep properties of the alloys. We calculated the relevance indicators using the

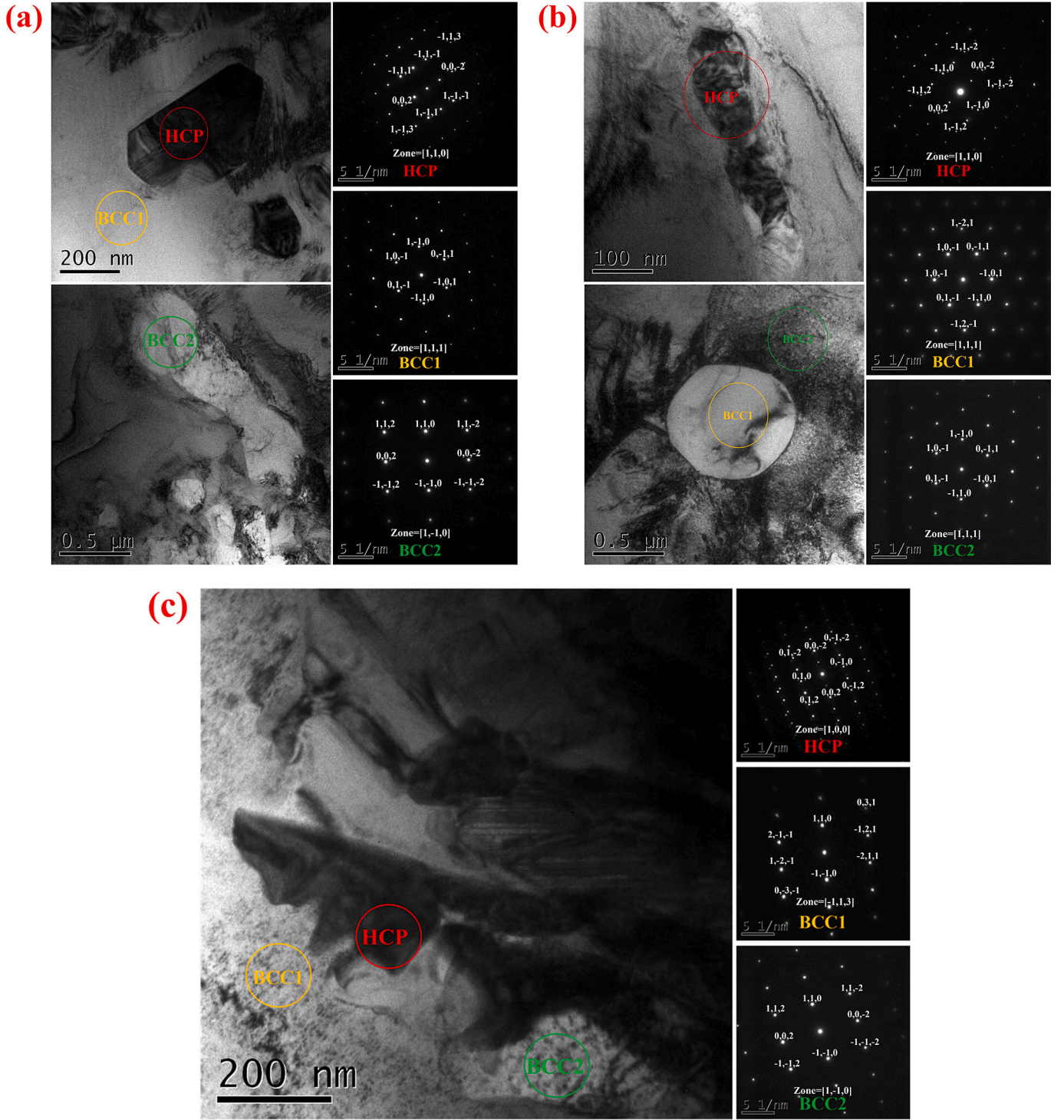


Fig. 6. TEM bright field and corresponding SAED images of (a) TiZrNbMoV (b) Ti_{1.5}ZrNbMoV, and (c) Ti₂ZrNbMoV.

following equations.

The creep displacement–time curves were fitted using the following empirical law [38].

$$h(t) = h_0 + a(t - t_0)^b + kt, \quad (4)$$

where $h(t)$ is the instantaneous indenter displacement, h_0 and t_0 are the indentation displacement and time at the beginning of creep, respectively, a , b , and k are the fitting constants, and t is the time. Table 3 lists the fitting constants. The correlation coefficient R^2 was >0.99 , indicating that the curves were fitted perfectly.

The strain rate $\dot{\epsilon}$ and applied stress σ for the Berkovich indenter are defined as follows [39,40]:

$$\dot{\epsilon} = \frac{1}{h} \frac{dh}{dt}, \quad (5)$$

$$\sigma = \frac{P}{24.56h^2} \quad (6)$$

The creep stress exponent n was calculated by [38,41]:

Table 1Chemical composition (at. %) of different phases of $\text{Ti}_x\text{ZrNbMoV}$ alloys by TEM-EDS.

	phase	Ti	Zr	Nb	Mo	V
TiZrNbMoV	Nominal	20.00	20.00	20.00	20.00	20.00
	HCP	7.17	33.95	8.73	20.79	29.34
	BCC1	24.00	35.07	16.87	10.35	13.69
$\text{Ti}_{1.5}\text{ZrNbMoV}$	BCC2	16.72	6.72	15.00	22.71	38.83
	Nominal	27.27	18.18	18.18	18.18	18.18
	HCP	7.91	35.13	9.48	18.71	28.75
	BCC1	26.14	17.27	17.31	19.21	20.05
$\text{Ti}_2\text{ZrNbMoV}$	BCC2	24.33	10.45	24.18	24.18	15.05
	Nominal	33.33	16.67	16.67	16.66	16.67
	HCP	21.99	59.39	5.96	6.05	6.58
	BCC1	32.29	24.67	14.89	13.27	14.86
	BCC2	33.19	9.54	17.36	17.11	22.78

Table 2Calculated parameters of VEC and δ for the $\text{Ti}_x\text{ZrNbMoV}$ RHEAs.

Alloy	VEC	δ (%)	ρ_{mix} (g/cm^3)	SYS ($\text{kPa}\cdot\text{m}^3/\text{kg}$)
TiZrNbMoV	4.8	5.78	7.13	312
$\text{Ti}_{1.5}\text{ZrNbMoV}$	4.73	5.51	6.89	245
$\text{Ti}_2\text{ZrNbMoV}$	4.67	5.28	6.69	237

$$n = \frac{\partial \ln \dot{\epsilon}}{\partial \ln \sigma} \quad (7)$$

The creep strain rate sensitivity m and creep stress exponent n are reciprocal to each other [40], as shown below:

$$m = \frac{1}{n} \quad (8)$$

Fig. 9(a)–(c) show the creep displacement–time curves, while Fig. 9(d) shows the creep depth of the $\text{Ti}_x\text{ZrNbMoV}$ alloys. Fig. 9(a)–(c) show that the creep behavior of the $\text{Ti}_x\text{ZrNbMoV}$ alloys can be divided into two stages. In the initial stage (transient creep stage), movable dislocation sources are activated; thus, the creep rate is fast, and the creep displacement increases rapidly. In the second stage (steady-state creep), the movable dislocation source is exhausted, and the dislocations pile up from the grain boundary. The second phase is hindered, resulting in a stable creep rate and slow creep displacement. As shown in Fig. 9(d), the creep depths in the whole creep stage increased from approximately 29.6 to 33 nm with increasing Ti content.

The steady-state creep stage slopes of $\ln(\dot{\epsilon})$ – $\ln(\sigma)$ curves were

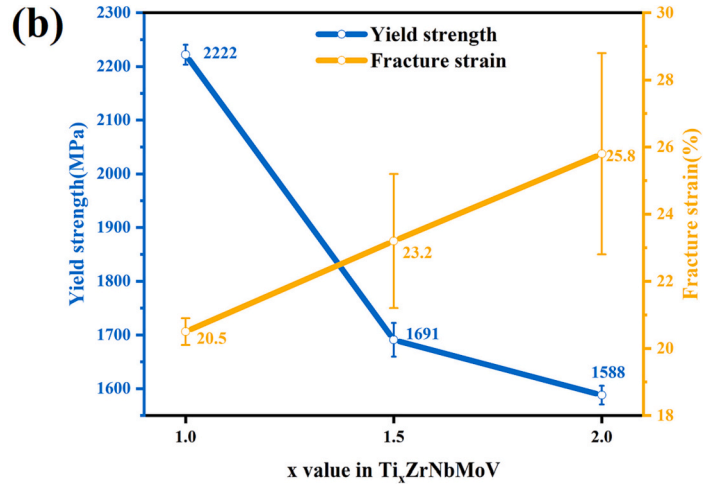
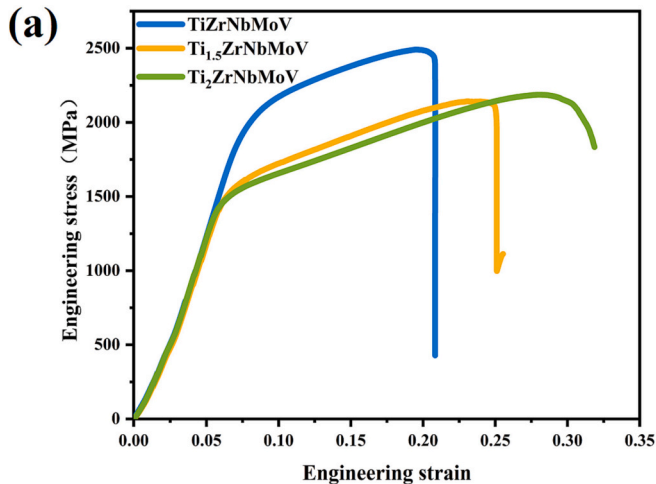


Fig. 7. (a) Engineering stress–strain curves of $\text{Ti}_x\text{ZrNbMoV}$ ($x = 1, 1.5$, and 2) RHEAs, (b) yield strength and fracture strain as functions of Ti concentration in $\text{Ti}_x\text{ZrNbMoV}$ ($x = 1, 1.5$, and 2) RHEAs.

calculated (by fitting) from the creep stress exponent n of the $\text{Ti}_x\text{ZrNbMoV}$ alloys (see Fig. 10(a)). The value of n decreased from 105 down to 72 and 34, respectively, as the Ti content increased.

The creep behavior is strongly correlated with the mechanism of grain boundary and dislocation movement in the creep process. The value of n may reflect the dislocation generation and annihilation mechanisms of the creep process [42]. The more dislocations involved, the more difficult the deformation, resulting in a higher n value [43]. Because $n > 3$, the dominant creep mechanism involved in all three $\text{Ti}_x\text{ZrNbMoV}$ alloys is considered to be dominated by dislocation climb [44]. The alloys' reduction in lattice distortion indicates a reduced

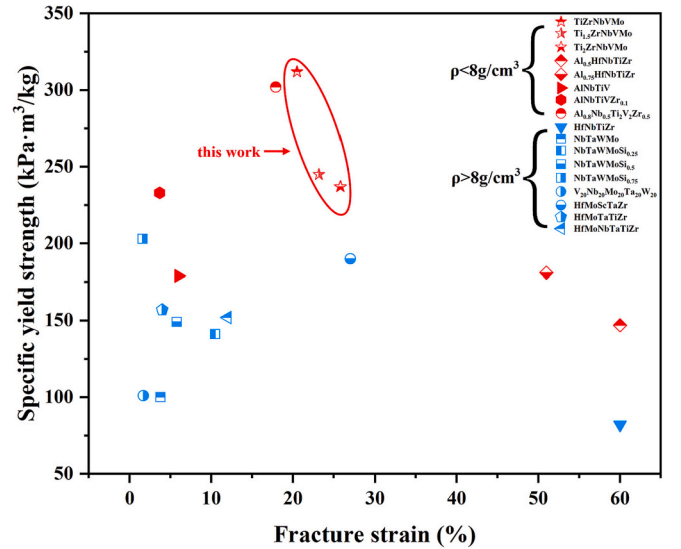


Fig. 8. Specific yield strength–fracture strain graph for the $\text{Ti}_x\text{ZrNbMoV}$ alloys and other reported RHEAs in the literature [4,30–35].

Table 3Fitting constants and R^2 of creep displacement–time curves of $\text{Ti}_x\text{ZrNbMoV}$ RHEAs.

Alloy	h_0	t_0	a	b	k	R^2
TiZrNbMoV	1820.462	199.867	10.195	0.347	−0.107	0.999
$\text{Ti}_{1.5}\text{ZrNbMoV}$	1868.752	199.862	13.201	0.293	−0.033	0.999
$\text{Ti}_2\text{ZrNbMoV}$	1685.844	200.067	6.910	0.435	−0.019	0.995

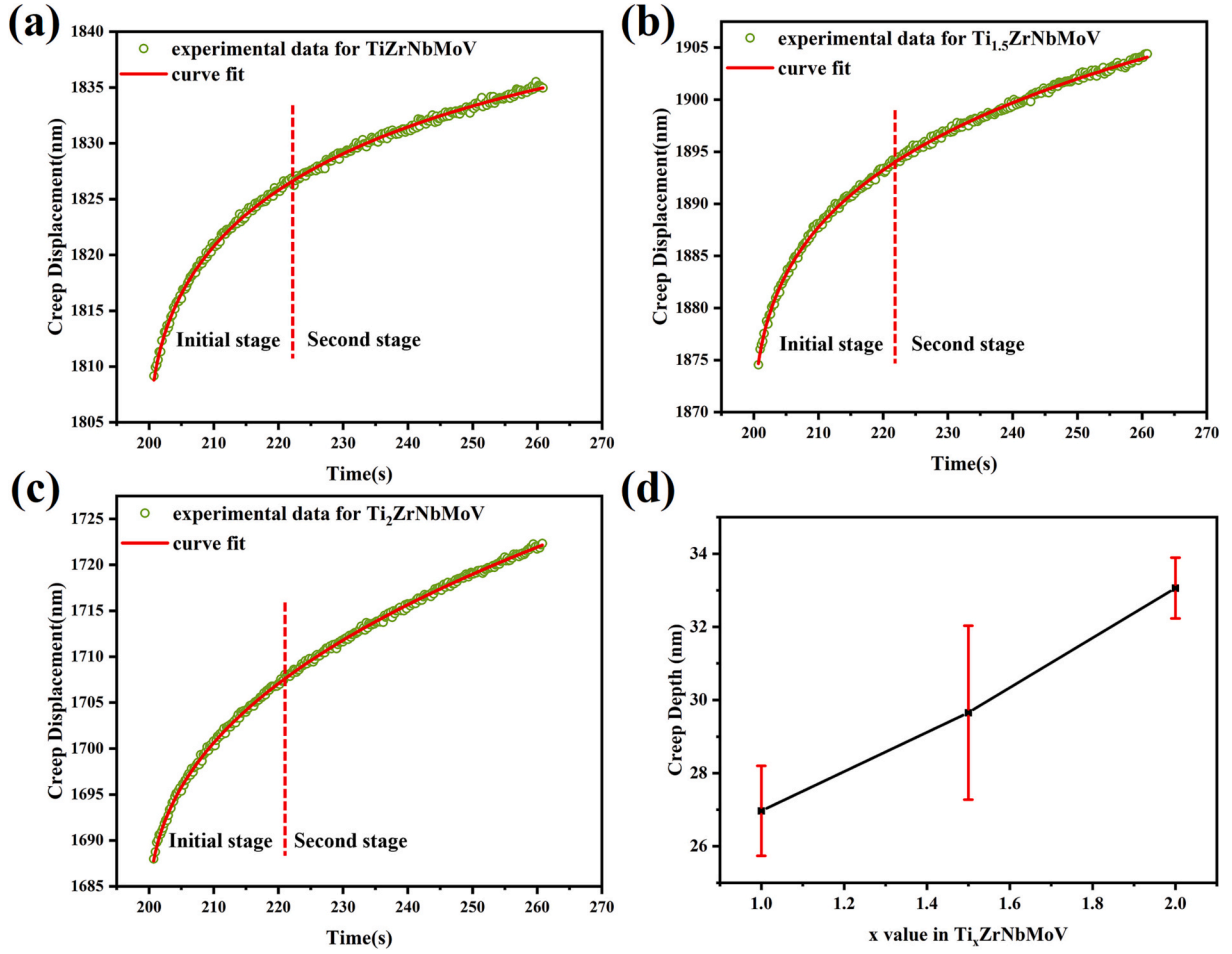


Fig. 9. Creep displacement–time curves of Ti_xZrNbMoV ($x = 1, 1.5$, and 2) RHEAs, (a) TiZrNbMoV, (b) Ti_{1.5}ZrNbMoV, and (c) Ti₂ZrNbMoV and (d) creep depth of Ti_xZrNbMoV ($x = 1, 1.5$, and 2) RHEAs.

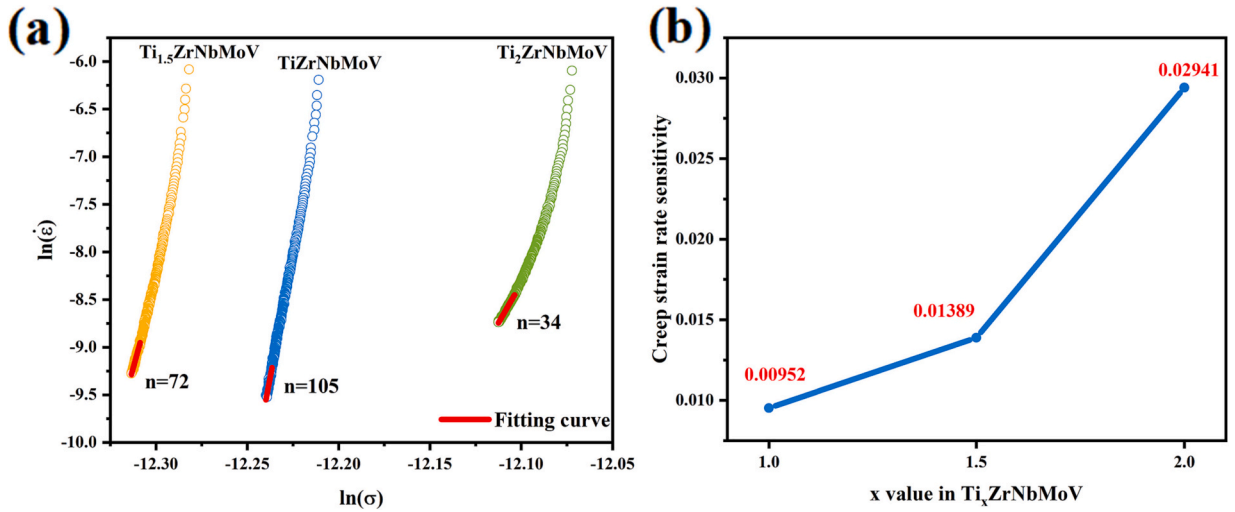


Fig. 10. (a) $\ln(\dot{\epsilon})$ – $\ln(\sigma)$ plot and (b) creep strain rate sensitivity of Ti_xZrNbMoV ($x = 1, 1.5$, and 2) RHEAs.

ability to prevent dislocation movement that results in a lower value of n . The decrease of creep stress exponent is also related to grain refinement. Coarser grains are favorable to the creep resistance of the alloys [45]. The creep rate increased with a decrease in grain size, resulting in a lower creep stress exponent. The grain size of single crystals can be regarded as infinite [46] with the best creep resistance. The

microstructure of an alloy also affects the value of n . Dendrites are considered more resistant to dislocation movement than equiaxed grains [41].

Fig. 10(b) shows the strain rate sensitivity of the Ti_xZrNbMoV RHEAs. The strain rate sensitivities m of the TiZrNbMoV, Ti_{1.5}ZrNbMoV, and Ti₂ZrNbMoV alloys were 0.00952, 0.01389, and 0.0294,

respectively. The strain rate sensitivity of the $\text{Ti}_x\text{ZrNbMoV}$ alloys is lower than that of some previously reported metallic materials [40,47–51]. The low strain-rate sensitivity of the $\text{Ti}_x\text{ZrNbMoV}$ RHEAs is mainly due to severe lattice distortion and second-phase nanocrystals. RHEAs have large differences in atomic radii, causing substantial lattice distortion and making components difficult to disperse, which helps them resist creep in the loading regime. Compared to single-phase entropy alloys, some of the second-phase nanocrystals of the $\text{Ti}_x\text{ZrNbMoV}$ RHEAs exist at the grain boundary, which provides more dislocations to prevent creep from continuing. In addition, HEAs with a BCC structure were found to have a lower strain-rate sensitivity [47].

4. Conclusions

In this study, we investigated the effects of varying the Ti content on the microstructures and mechanical properties of $\text{Ti}_x\text{ZrNbMoV}$ ($x = 1, 1.5, 2$) RHEAs. Based on the results of our study, we present the following conclusions:

- (1) The $\text{Ti}_x\text{ZrNbMoV}$ alloys mainly comprised a BCC phase and a few secondary HCP phases. Second-phase nanocrystals existed primarily at the grain boundaries of the BCC phase. The proportion of the second phase decreased with increasing Ti content, and the microstructure of the $\text{Ti}_x\text{ZrNbMoV}$ alloys underwent dendritic to equiaxed-dendritic transformation. Meanwhile, the average grain size of RHEAs decreased from 81 to 16 μm with increased Ti content.
- (2) The $\text{Ti}_x\text{ZrNbMoV}$ alloys possessed excellent mechanical properties, as well as a good balance between strength and ductility, low density (6.69–7.13 g/cm^3), and high SYS (237–312 $\text{kPa}\cdot\text{m}^3/\text{kg}$). The high strength of the $\text{Ti}_x\text{ZrNbMoV}$ alloys is attributed mainly to solution strengthening and second-phase strengthening mechanisms. The reduction in the degree of lattice distortion and the secondary phase resulted in a decrease in strength. In addition, the grain refinement of RHEAs gave rise to an improvement in ductility with an increase in Ti content.
- (3) The strain rate sensitivity of the RHEAs was 0.00952–0.02941 at the steady-state creep stage in the nanoindentation creep test. The low strain-rate sensitivity of the $\text{Ti}_x\text{ZrNbMoV}$ RHEAs was mainly due to severe lattice distortion and second-phase nanocrystals.

CRedit authorship contribution statement

Zhenghang Cao: Conceptualization, Formal analysis, Investigation, Methodology, Software, Writing – original draft. **Huaican Chen:** Data curation, Writing – original draft. **Wei Wang:** Formal analysis, Investigation, Methodology. **Jianwen Guo:** Funding acquisition, Writing – review & editing. **Zhenzhong Sun:** Funding acquisition, Project administration, Supervision. **Yang Hai:** Conceptualization, Funding acquisition, Resources, Supervision, Writing – review & editing. **Wen Yin:** Conceptualization, Funding acquisition, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declared that they have no conflicts of interest to this work.

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

Data availability

Data will be made available on request.

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